CAD Readers and Data Exchange

As of JobLine G-Code for VS Code 0.3.12

Models and setup data

CAD System / Data	Format	Version	Read	Write	Feature / Data Exchange	Sketch / 2D Data
STEP	*.stp, *.step	AP203 / AP214 / AP242 (ISO 10303-21)	✓[1]			
STL	*.stl	ASCII and binary	✓			
Parasolid (text)	*.x_t, *.xmt_txt, *.xmt	Text transmit files	✓[2]			
Parasolid (binary)	*.x_b	—	[3]			
Autodesk Fusion setup	*.f3d	Fusion archive (ZIP)	✓[4]		✓[4]	
JobLine machine	*.jblmachine (JSON)	Format 1.0	✓	✓	✓[5]	
RevPack	revpack.json	`jobline.revpack/2`	[6]		[6]	
RevPack (v1 manifest)	manifest JSON	v1 (gcode + goal models)	[7]			
Lathe insert codes	ISO designation	Shapes C, D, R, S, T, V, W	✓			

[1] Tessellated through OpenCascade (occt-import-js 0.0.23, at most 300,000 triangles) and shown as a scene or target model. The schema name comes from FILE_SCHEMA. Features, PMI and assembly structure are not read.

[2] Bounds only: every 3D point in the file is collected and the bounding envelope is drawn. Surfaces are not rebuilt.

[3] The import dialog lists *.x_b, but the file is read as text, so binary Parasolid is not decoded. Treat it as not supported and export *.x_t or STEP.

[4] Reads the active setup's stock size and allowances, work offset, WCS orientation and origin, operation names, the model envelope and the preview PNG. The solid model and toolpaths are not read.

[5] `.jblmachine` is byte-identical in the extension, jobline.ai-CAM and the JobLine.ai platform (`src/presets/jblMachine.ts`), so a machine exported from one opens in the others.

[6] In draft PR #7, not released (0.3.14 on that branch): RevPack v2 loads target, stock, workholding, tools and machine. STEP targets go through OpenCascade and are cross-checked against the paired STL. Its AR4 real-part checks compare STEP and STL digests from local, gitignored Annin AR4 files.

[7] Development harness only (`npm run visualizer:harness:revpack`), not a user command.

CAD Readers and Data Exchange (continued)

G-code and control dialects

JobLine's equivalent of a CAD reader: the program formats the extension opens, checks and explains. Any G-code file also gets highlighting, the toolpath visualizer, the explainer and the Toolbox edits.

Control	Program files	Models named in the app	Read	Validate	Post	Dialect-specific parsing
Fanuc	*.nc, *.tap, *.sub, *.lib	0i / 30i / 31i	✓	✓[8]	[9]	Macro B variables and expressions (common #100–#999, system #1000+ read-only)
Haas	*.nc, *.5ax, *.ncc, *.ncp, *.ncf, *.nc1	NGC	✓	✓[8]	[9]	Fanuc-style. Haas Classic is not handled separately
Siemens Sinumerik	*.mpf, *.spf	840D / 828D	✓	✓[8]	[9]	CYCLExx, MCALL, HOLES1/2, POCKET1–4, SLOT1/2, LONGHOLE, SWIVEL; R-parameters
Okuma OSP	*.min, *.prg	OSP-P	✓	✓[8]	[9]	$ program headers, #V variables, CALL OBn; lathe detection
Mazak	*.eia, *.iso	Smooth / Matrix (EIA/ISO)	✓	✓[8]	[9]	EIA/ISO G-code only. Mazatrol conversational is not read
Fadal	*.fgc	—	✓	✓[8]	[9]	Generic rules
Generic / LinuxCNC	*.gcode, *.ngc, *.g, *.cnc, *.dnc	—	✓	✓[8]	[9]	Generic rules
Heidenhain TNC	*.cyc, *.tab	—	[10]			Control family recognised in .jblmachine only; Klartext not parsed
Mitsubishi	—	—	[10]			Control family recognised in .jblmachine only
Fanuc Robot	*.ls, *.tp	TP / LS	✓[11]			Language and control entry; no robot kinematics
ABB RAPID	*.mod, *.pgf	RAPID	✓[11]			Language and control entry; no robot kinematics
KUKA KRL / CL data	*.src, *.ptp, *.apt, *.cls	—	[12]			Opened as G-code; KRL and CL/APT are not interpreted

[8] One generic ISO/Fanuc-style rule set for every control: canned-cycle Z/R/Q, arc validity, spindle and coolant still on at program end, and axis travel against the active machine preset. The chosen dialect changes tokenising, explanations and lathe detection. Generic tests don't certify every dialect.

[9] The extension has no post-processor. It edits existing programs (formatter, renumbering, feed/speed scaling, axis shifts, safety header). Posting is in jobline.ai-CAM (Haas NGC mill).

[10] Listed in the shared control-family list but not offered in **Select CNC / Robot Control Type**. *.h / *.hpp appear in `fileTypes.ts` but not in the language contribution.

[11] Syntax highlighting (shared G-code grammar) and a control entry for preset matching only.

[12] Registered as G-code extensions, so they get highlighting and the generic G-code reader only.

⚠ **Notice: Not in this release: tool-library import (JSON/CSV) and CAM connections (ESPRIT, Mastercam, GibbsCAM, Hexagon, Fusion API, Cura). Those classes are stubs that return nothing. DXF, IGES, SolidWorks and other native CAD formats are not read.**